

Q1 2021 (31 Aug Shift 2)

If $\int \frac{\sin x}{\sin^3 x + \cos^3 x} dx =$

$$\alpha \log_c |1 + \tan x| + \beta \log_e |1 - \tan x + \tan^2 x| + \gamma \tan^{-1} \left(\frac{2 \tan x - 1}{\sqrt{3}} \right) + C,$$

when C is constant of integration, then the value of $18(\alpha + \beta + \gamma^2)$ is .

Q2 2021 (31 Aug Shift 1)

If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt$, $x > -2$, and $\phi(0) = 4$ then $\phi(2)$ is.

Q3 2021 (31 Aug Shift 1)

The integral $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}} dx$ is equal to :

(where C is a constant of integration)

(1) $\frac{3}{4} \left(\frac{x+2}{x-1} \right)^{\frac{1}{4}} + C$

(2) $\frac{3}{4} \left(\frac{x+2}{x-1} \right)^{\frac{5}{4}} + C$

(3) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{1}{4}} + C$

(4) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{\frac{5}{4}} + C$

Q4 2021 (27 Aug Shift 2)

If $\int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx = \frac{1}{14}(ux + v \log_e(4e^x + 7e^{-x})) + C,$

where C is a constant of integration, then $u + v$ is equal to .

Q5 2021 (27 Aug Shift 1)

If $\int \frac{dx}{(x^2 + x + 1)^2} = a \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + b \left(\frac{2x+1}{x^2 + x + 1} \right) + C$ $x > 0$ where C is the constant of integration, then the

value of $9(\sqrt{3}a + b)$ is equal to .

Answer Key

Q1 (3)

Q2 (4)

Q3 (3)

Q4 (7)

Q5 (15)

Q1 (3)

$$= \int \frac{\frac{\sin x}{\cos^3 x}}{1 + \tan^3 x} dx = \int \frac{\tan x \cdot \sec^2 x}{(\tan x + 1)(1 + \tan^2 x - \tan x)} dx$$

$$\text{Let } \tan x = t \Rightarrow \sec^2 x \cdot dx = dt$$

$$= \int \frac{t}{(t+1)(t^2-t+1)} dt$$

$$= \int \left(\frac{A}{t+1} + \frac{B(2t-1)}{t^2-t+1} + \frac{C}{t^2-t+1} \right) dx$$

$$\Rightarrow A(t^2 - t + 1) + B(2t - 1)(t^2 - t + 1) + C(t + 1) = t$$

$$\Rightarrow t^2(A + 2B) + t(-A + B + C) + A - B + C = 1$$

$$\therefore A + 2B = 0$$

$$-A + B + C = 1$$

$$A - B + C = 0$$

$$\Rightarrow C = \frac{1}{2} \Rightarrow A - B = -\frac{1}{2} \quad \dots (4)$$

$$A + 2B = 0$$

$$A - B = -\frac{1}{2}$$

$$\Rightarrow 3B = \frac{1}{2} \Rightarrow B = \frac{1}{6}$$

$$A = -\frac{1}{3}$$

$$I = -\frac{1}{3} \int \frac{dt}{1+t} + \frac{1}{6} \int \frac{2t-1}{t^2-t+1} dt + \frac{1}{2} \int \frac{dt}{t^2-t+1}$$

$$= -\frac{1}{3} \ln|(1 + \tan x)| + \frac{1}{6} \ell |\tan^2 x - \tan x + 1|$$

$$+ \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\left(\tan x - \frac{1}{2} \right)}{\frac{\sqrt{3}}{2}} \right)$$

$$= -\frac{1}{3} \ln|(1 + \tan x)| + \frac{1}{6} \ell |\tan^2 x - \tan x + 1|$$

$$+ \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2 \tan x - 1}{\sqrt{3}} \right) + C$$

$$\alpha = -\frac{1}{3}, \beta = \frac{1}{6}, \gamma = \frac{1}{\sqrt{3}}$$

$$18(\alpha + \beta + \gamma^2) = 18 \left(-\frac{1}{3} + \frac{1}{6} + \frac{1}{3} \right) = 3$$

Q2 (4)

$$x\phi(x) = \int_5^x 3t^2 - 2\phi'(t)dt$$

$$x\phi(x) = x^3 - 125 - 2[\phi(x) - \phi(5)]$$

$$x\phi(x) = x^3 - 125 - 2\phi(x) - 2\phi(5)$$

$$\phi(0) = 4 \Rightarrow \phi(5) = -\frac{133}{2}$$

$$\phi(x) = \frac{x^3+8}{x+2}$$

$$\phi(2) = 4$$

Q3 (3)

$$\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}}$$

$$= \int \frac{dx}{\left(\frac{x+2}{x-1}\right)^{5/4} \cdot (x-1)^2}$$

$$\text{put } \frac{x+2}{x-1} = t$$

$$= -\frac{1}{3} \int \frac{dt}{t^{5/4}}$$

$$= \frac{4}{3} \cdot \frac{1}{t^{1/4}} + C$$

$$= \frac{4}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + C$$

Q4 (7)

$$\int \frac{2e^x}{4e^x+7e^{-x}}dx + 3 \int \frac{e^{-x}}{4e^x+7e^{-x}}dx$$

$$= \int \frac{2e^{2x}}{4e^{2x}+7}dx + 3 \int \frac{e^{-2x}}{4+7e^{-2x}}dx$$

$$\text{Let } 4e^{2x} + 7 = T$$

$$8e^{2x} dx = dT$$

$$2e^{2x} dx = \frac{dT}{4}$$

$$\text{Let}$$

$$4 + 7e^{-2x} = t$$

$$-14e^{-2x} dx = dt$$

$$e^{-2x} dx = -\frac{dt}{14}$$

$$\int \frac{dT}{4T} - \frac{3}{14} \int \frac{dt}{t}$$

$$= \frac{1}{4} \log T - \frac{3}{14} \log t + C$$

$$= \frac{1}{4} \log(4e^{2x} + 7) - \frac{3}{14} \log(4 + 7e^{-2x}) + C$$

$$= \frac{1}{14} \left[\frac{1}{2} \log(4e^x + 7e^{-x}) + \frac{13}{2} x \right] + C$$

$$u = \frac{13}{2}, v = \frac{1}{2} \Rightarrow u + v = 7$$

Aliter :

$$2e^x + 3e^{-x} = A(4e^x + 7e^{-x}) + B(4e^x - 7e^{-x}) + \lambda$$

$$2 = 4A + 4B; 3 = 7A - 7B; \lambda = 0$$

$$A + B = \frac{1}{2}$$

$$A - B = \frac{3}{7}$$

$$A = \frac{1}{2} \left(\frac{1}{2} + \frac{3}{7} \right) = \frac{7+6}{28} = \frac{13}{28}$$

$$B = A - \frac{3}{7} = \frac{13}{28} - \frac{3}{7} = \frac{13-12}{28} = \frac{1}{28}$$

$$\int \frac{13}{28} dx + \frac{1}{28} \int \frac{4e^x - 7e^{-x}}{4e^x + 7e^{-x}} dx$$

$$\frac{13}{28}x + \frac{1}{28} \ln |4e^x + 7e^{-x}| + C$$

$$u = \frac{13}{2}; v = \frac{1}{2}$$

$$\Rightarrow u + v = 7$$

Q5 (15)

$$I = \int \frac{dx}{\left[\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right]^2}$$

$$\int \frac{dt}{\left(t^2 + \frac{3}{4} \right)^2} \quad \left(\text{Put } x + \frac{1}{2} = t \right)$$

$$= \frac{\sqrt{3}}{2} \int \frac{\sec^2 \theta d\theta}{\frac{9}{16} \sec^4 \theta} \quad \left(\text{Put } t = \frac{\sqrt{3}}{2} \tan \theta \right)$$

$$= \frac{4\sqrt{3}}{9} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{4\sqrt{3}}{9} \left[\theta + \frac{\sin 2\theta}{2} \right] + c$$

$$= \frac{4\sqrt{3}}{9} \left[\tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{\sqrt{3}(2x+1)}{3+(2x+1)^2} \right] + c$$

$$= \frac{4\sqrt{3}}{9} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{1}{3} \left(\frac{2x+1}{x^2+x+1} \right) + c$$

$$\text{Hence, } 9(\sqrt{3}a + b) = 15$$